



A CAPSTONE PROJECT REPORT

MEMORY ALLOCATION SIMULATOR WITH FRAGMENTATION ANALYSIS

**CSA0401 – OPERATING SYSTEM
MODULE 3 – MEMORY VISUALIZATION**

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "MEMORY ALLOCATION SIMULATOR WITH FRAGMENTATION ANALYSIS" has been carried out by NITHISH KUMAR S (Reg. No. 1924260144) under the supervision of Dr. Vinoth D and is submitted in partial fulfilment of the requirements for CSA0401 – OPERATING SYSTEM at SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I AM, NITHISH KUMAR S, Registration No. 1924260144, hereby declare that the Capstone Project Work entitled "MEMORY ALLOCATION SIMULATOR WITH FRAGMENTATION ANALYSIS" is my own bonafide academic work prepared for CSA0401 – OPERATING SYSTEM, with emphasis on MODULE 3 – MEMORY VISUALIZATION. External resources and tools have been appropriately acknowledged.

INDIVIDUAL CONTRIBUTION STATEMENT

Student	Responsibility	Contribution
NITHISH KUMAR S / 1924260144	Module 3 visualization, testing, integration and report	100%

ABSTRACT

Memory management is a fundamental responsibility of an operating system because processes must be assigned memory efficiently while free regions are tracked and reused. This capstone project provides an interactive environment for simulating First Fit, Best Fit, Worst Fit and Next Fit. Module 3 – Memory Visualization converts allocation results into an easy-to-read memory map showing allocated/free regions, process ID and memory size, block boundaries, the current algorithm and utilization. For the supplied workload, Best Fit allocates all four processes with 100% success and 86.9% utilization, while the other three strategies show 75% success and 52.5% utilization. The project connects OS theory with observable behaviour and measurable analysis.

Keywords: Operating System, Memory Allocation, Memory Visualization, Fragmentation, First Fit, Best Fit

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CHAPTER 1: INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Memory management allocates RAM to processes, releases memory safely and maintains occupied and free regions. Allocation strategy affects speed, utilization and fragmentation.

1.2 Need for the Project

Students often learn allocation algorithms through static examples. Poor allocation can create unused memory and fragmented free space, while manual comparison of multiple algorithms is time-consuming. An interactive simulator makes the relationship between algorithm logic

and visible memory behaviour easier to understand.

1.3 Problem Statement

Design and implement an interactive memory-allocation simulator that converts allocation decisions into a clear visual representation of occupied and free memory regions while supporting fragmentation and utilization analysis.

1.4 Project Aim

To develop a web-based memory allocation simulator that makes operating-system memory allocation behaviour easy to visualize, analyze and compare.

1.5 Project Objectives

1. Simulate allocation. 2. Implement First Fit, Best Fit, Worst Fit and Next Fit. 3. Visualize allocated and free memory. 4. Analyze fragmentation. 5. Compare performance.

1.6 Scope

The project includes process and memory-block input, allocation simulation, memory-map visualization and performance metrics. It excludes real kernel memory control and production-grade process scheduling. The simulator treats memory blocks as abstract regions.

1.7 Expected Engineering Outcome

An educational prototype that transforms allocation results into a visual memory map and measurable performance indicators.

CHAPTER 2: LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The review is centred on memory management, allocation strategies, fragmentation and visualization for operating-systems education, using the supplied project presentation as the primary source for the project modules and sample results.

2.2 Review of Existing Work

Traditional teaching uses worked examples, tables and diagrams. These explain allocation rules but do not provide the same immediate interaction as a simulator. The proposed system combines the allocation engine with a visual memory map and measurable metrics.

2.3 Comparative Analysis

Static examples are simple but limited; manual tables are useful but time-consuming; text-only simulation is compact but visually weak; the proposed interactive simulator combines simulation, visualization and analysis.

2.4 Research / Engineering Gap

The project addresses the integration gap between allocation decisions and immediate visual representation. It is not presented as a new allocation algorithm; its contribution is the interactive educational workflow.

2.5 Proposed Contribution

The workflow is Input → Allocation → Visualization → Fragmentation Analysis → Performance Comparison → Report. Module 3 provides the visual bridge between allocation and analysis.

CHAPTER 3: ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Functional requirements include valid input, allocated/free display, PID and size labels, block boundaries, algorithm identification and utilization. Non-functional requirements include readability and interactive updates.

3.2 Constraints & Applicable Standards

The project is a software-only academic prototype. Main constraints are project time, correctness, interface readability and educational scope. No physical safety standard directly applies because the prototype does not control physical memory.

3.3 Design Specifications

Positive process and block sizes, identifiable IDs, clear allocated/free state, visible PID and size, block boundaries and utilization summary.

3.4 Alternative Solutions & Evaluation

Alternative 1 is text-only simulation, which is simple but less visual. Alternative 2 is an interactive web visualization, which requires more development but better supports the assigned visualization objective. The interactive design is selected.

3.5 Selected Design & Detailed Design

Architecture: User → Process & Memory Input → Allocation Engine → Memory Visualization → Fragmentation Analysis → Performance Comparison → Report. Module 3 receives allocation results and renders occupied/free regions with process information, boundaries, algorithm and utilization.

3.6 Trade-off Analysis

A visual memory map is preferred over text-only output because it improves immediate understanding. A Web UI + C simulation core is used to separate presentation from allocation logic. Metric summaries are retained because they make the comparison measurable.

3.7 Sustainability, Ethics, Safety, Risk & Security

The software-only design avoids specialized hardware. Academic integrity requires acknowledgement of project resources. Input validation reduces invalid-state risk; cross-checking allocation results against the visualization reduces display errors. The main risks are invalid inputs, integration mismatch and visual overcrowding.

CHAPTER 4: IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The project follows a modular workflow: input preparation, allocation, visualization, fragmentation analysis and performance comparison. Resources identified in the supplied presentation are HTML, CSS, JavaScript, C and Visual Studio Code.

4.2 Software Implementation & Modern Tools Used

HTML provides interface structure; CSS supports UI layout and memory-map presentation;

JavaScript provides validation, simulation controls and live updates; C provides allocation algorithms and metric calculations; Visual Studio Code supports development and debugging.

4.3 Test Plan & Verification

Tests verify allocated/free regions, PID and size labels, block boundaries, current algorithm identification and utilization. A test passes when the visualization agrees with the allocation result and the displayed metric matches the supplied simulation result.

4.4 Results

Representative workload: P1=90 KB, P2=210 KB, P3=120 KB, P4=275 KB; blocks B1=100 KB, B2=250 KB, B3=300 KB and B4=150 KB.

4.5 Data Analysis & Engineering Judgment

For the supplied workload, Best Fit produces the strongest result: all four processes are allocated, internal fragmentation is 105 KB, external fragmentation is 0 KB, success is 100% and utilization is 86.9%. First Fit, Worst Fit and Next Fit each show 75% success and 52.5% utilization in the supplied results.

4.6 Comparison with Existing Solutions & Achievement of Objectives

The supplied project objectives are achieved: four allocation strategies are represented, memory is visualized, fragmentation is quantified and algorithm performance is compared.

4.7 Strengths & Limitations

Strengths: easy visualization, interactive simulation, clear comparison, useful for OS learning and viva demonstrations. Limitations: educational prototype scope, possible visual crowding for large workloads, and limited supplied sample workload.

4.8 Project Planning, Roles & Budget

NITHISH KUMAR S is responsible for the individual project contribution (100%). Milestones: problem definition; algorithm/module design; Module 3 visualization; testing; report preparation. No specialized hardware is required.

Algorithm	Allocation		Internal	External	Success	Utilization
First Fit	P1→B1, P2→B2, P3→B3		230 KB	150 KB	75%	52.5%
Best Fit	P1→B1, P2→B2, P3→B4, 3 P4→B		105 KB	0 KB	100%	86.9%
Worst Fit	P1→B3, P2→B2, P3→B4		280 KB	100 KB	75%	52.5%
Next Fit	P1→B1, P2→B2, P3→B3		230 KB	150 KB	75%	52.5%

CHAPTER 5: CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The simulator connects operating-system theory with observable memory behaviour. Module 3 converts allocation results into an easy-to-read memory map showing occupied/free regions, process information, boundaries and utilization. The sample results demonstrate that allocation strategy can significantly change memory outcomes.

5.2 Future Work

Future enhancements identified in the supplied project are real-time process allocation, advanced visualization and animations, more allocation algorithms, automatic workload generation and exportable performance reports.

5.3 Professional Learning & Individual Reflection

NITHISH KUMAR S developed practical knowledge of allocation strategies, memory visualization, fragmentation metrics, modular design and testing. The main challenge was making an abstract allocation process visually understandable. The map-based design was selected because it directly shows where memory is used and where free space remains. If repeated, the project could be extended with larger workloads, animation and exportable reports.

REFERENCES

- [1] "Memory Allocation Simulator with Fragmentation Analysis," Capstone Project Presentation, CSA 0401 – Operating System, SIMATS Engineering, 2026.
- [2] "Capstone Project Report - New Format - Sep 2026," SIMATS Engineering, 2026.
- [3] HTML, CSS, JavaScript, C and Visual Studio Code, as identified in the supplied project presentation.

APPENDICES

Appendix A – Detailed Sample Workload

Item	Value
Processes	P1=90 KB, P2=210 KB, P3=120 KB, P4=275 KB
Blocks	B1=100 KB, B2=250 KB, B3=300 KB, B4=150 KB
Best Fit	All 4 allocated; 105 KB internal; 0 KB external; 100% success; 86.9% utilization

Appendix B – Module 3 Visual Representation

OS → P1 (90 KB) → P2 (210 KB) → FREE (40 KB) → P3 (120 KB) → FREE (150 KB)

Appendix C – Module Mapping

Module	Purpose
Module 1	Process & Memory Input Management

Module 2	Memory Allocation Engine
Module 3	Memory Visualization – assigned module
Module 4	Fragmentation Analysis
Module 5	Performance Analysis & Comparison

CAPSTONE PROJECT OUTCOME MAPPING SHEET

Outcome	SO / Area	Evidence
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3.7
Experimentation and engineering judgment	SO6	Ch. 4.3–4.6
Independent knowledge acquisition	SO7	Ch. 5.3
Integrated/system-level engineering	SO1/SO2	Ch. 3.5
Project planning and management	SO5	Ch. 4.8
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4.8